UNIT 5

ROCKS

Learning objectives:

- Identify the three main types of rocks: igneous, sedimentary, and metamorphic, and understand their formation.
- Differentiate between intrusive and extrusive igneous rocks with examples like granite and basalt.
- Explain the rock cycle and how rocks change from one type to another over time.
- Describe the characteristics and uses of sedimentary and metamorphic rocks, including fossils in sedimentary rocks and marble in metamorphic rocks.

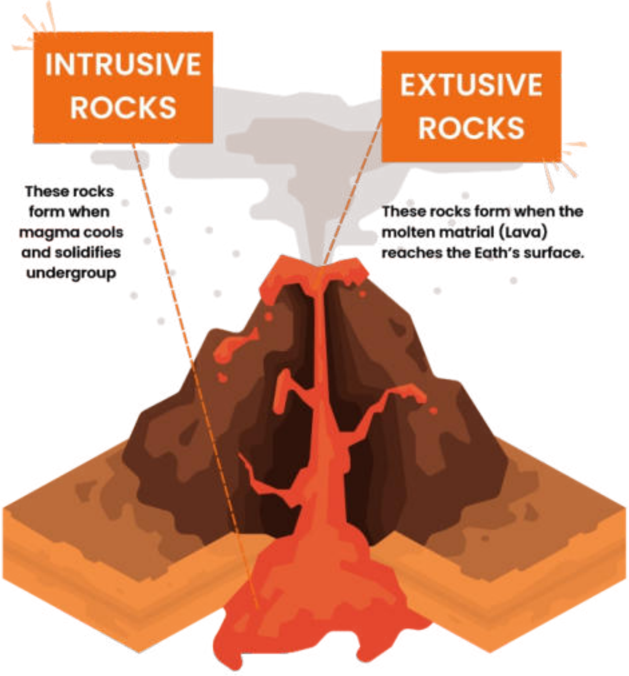
ROCKS

Rocks are the solid materials that make up the Earth's crust. They are found everywhere on Earth—in mountains, valleys, and even underwater. Rocks are made of minerals, and their properties depend on the type and arrangement of these minerals. Rocks are divided into three main types based on how they are formed: igneous, sedimentary, and metamorphic rocks.

IGNEOUS ROCKS

Igneous rocks form when molten rock (magma or lava) cools and solidifies.

- **Intrusive Igneous Rocks**: Formed when magma cools slowly beneath the Earth's surface, creating large crystals.
  Example: Granite.
- **Extrusive Igneous Rocks**: Formed when lava cools quickly on the surface, creating small crystals or a smooth texture.
  Example: Basalt.



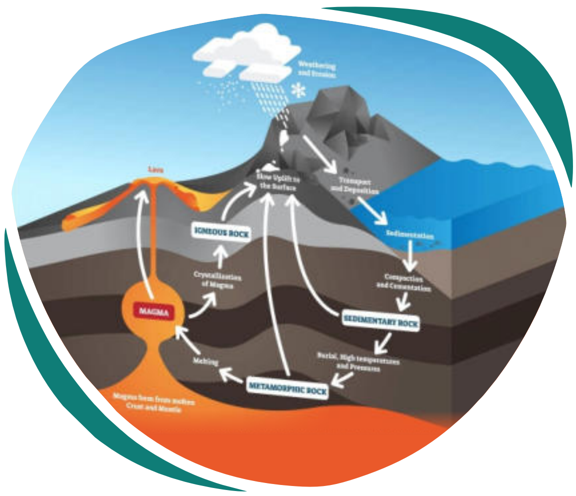
SEDIMENTARY ROCKS

Sedimentary rocks are formed from small pieces of other rocks, sand, and organic material that are compressed and cemented over time. They often have layers and may contain fossils. Example: Sandstone, limestone.

METAMORPHIC ROCKS

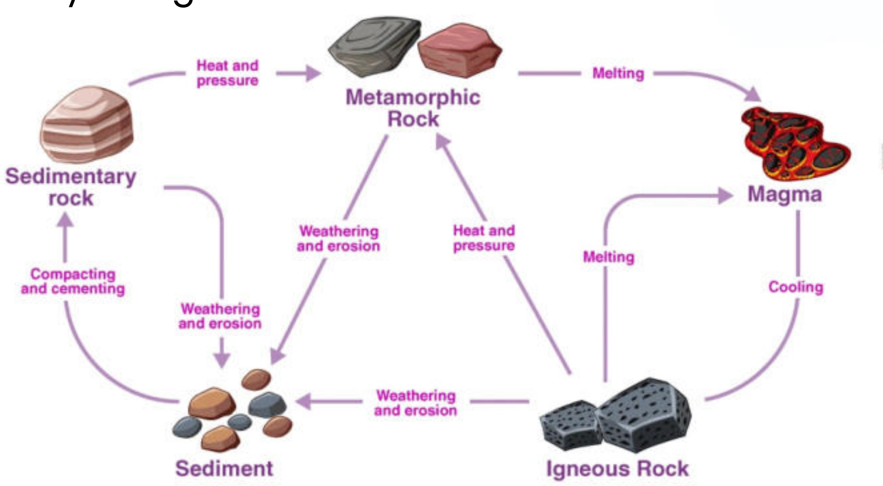
Metamorphic rocks are formed when existing rocks are changed by heat and pressure deep within the Earth. These rocks often have a banded or crystalline texture.

Example: Marble, slate.



ROCK CYCLE

Rocks constantly change from one type to another in a process called the rock cycle. Igneous rocks can break down to form sedimentary rocks, and both types can be transformed into metamorphic rocks. Metamorphic rocks can melt to form magma, starting the cycle again.



Fill in the blanks.

1. Rocks are made up of ________________ .

2. ________________ rocks are formed when magma cools and solidifies.

3. Fossils are often found in ________________ rocks.

4. Marble is an example of a ________________ rock.

5. The ________________ is a process where rocks change from one type to another.

6. ________________ igneous rocks form beneath the Earth's surface.

7. Sandstone is a type of ________________ rock.

8. Heat and ________________ create metamorphic rocks.

Write “T” for True or “F” for False.

1. Igneous rocks are formed by heat and pressure.

2. Sedimentary rocks often have layers.

3. Metamorphic rocks are formed by the cooling of lava.

4. Granite is an intrusive igneous rock.

5. The rock cycle shows how rocks change over time.

6. Fossils are only found in metamorphic rocks.

7. Basalt is an extrusive igneous rock.

8. Limestone is a type of sedimentary rock.

Choose the correct answer.

1. Which rock type forms from cooling lava?

- Sedimentary
- Igneous
- Metamorphic
- Fossil

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2. What type of rock is marble?

- Igneous
- Sedimentary
- Metamorphic
- Intrusive

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3. Rocks that form layers are called:

- Igneous
- Sedimentary
- Metamorphic
- Magma

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4. What process changes rocks into metamorphic rocks?

- Cooling
- Heating and pressure
- Erosion
- Melting

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5. Granite is an example of:

- Extrusive igneous rock
- Intrusive igneous rock
- Sedimentary rock
- Metamorphic rock

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6. Which rock contains fossils?

- Marble
- Sandstone
- Basalt
- Granite

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7. The rock cycle describes:

- The weathering of rocks
- The formation and transformation of rocks
- The melting of magma
- The layering of sediment

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8. What forms sedimentary rocks?

- Melting
- Cementation of small particles
- Pressure only
- Cooling magma

Answer the following questions.

Q1: What are the three types of rocks, and how are they formed?

Q2: How are intrusive and extrusive igneous rocks different?

Q3: Describe the process of the rock cycle.

Q4: Define rocks.

Q5: Why are fossils mostly found in sedimentary rocks?

Critical Thinking

If the rock cycle stopped, how would it affect the formation of new rocks and landscapes?

Encourage students to discuss the role of natural processes like erosion, pressure, and heat in shaping Earth's surface.

Rock Identification Activity

ACTIVITY

Identify and classify rocks based on their properties.

- **Materials Needed**: A variety of rock samples (granite, basalt, sandstone, marble), a magnifying glass, and a worksheet.
- **Instructions**:

1. Examine each rock with a magnifying glass.
2. Observe its texture, colour, and layers.
3. Classify each rock as igneous, sedimentary, or metamorphic on the worksheet.

- **Outcome**: Students will learn to differentiate between rock types based on observable characteristics.

Fun Fact

Marble, a metamorphic rock, is often used in making statues and buildings.